Code No: 117AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, February/March - 2022
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. What is state space tree? Explain with an example. [15]

2. 8-queens problem seeks to place 8-queens in an 8 × 8 chessboard such that no two queens attack each other. Formulate this problem as a constrained satisfaction problem. [15]

3. Consider the following set of axioms
   a) Everyone who loves someone who loves them back is happy
   b) Mary loves everyone
   c) There is someone who loves Mary
   From the above statements conclude that: Mary is happy. [15]

4. Describe how a Knowledge Representation and Reasoning system is built with a semantic network processing system. [15]

5. What is an expert system? Briefly describe five major components of an expert system. Using a suitable query, explain the working of an inference engine in a rule-based expert system. [15]

6. Explain in detail about the concept of uncertainty using Bayesian belief network. [15]

7. With the help of example, explain how classification is performed using decision trees. [15]

8. Explain in detail about the concept of semantic analysis in natural language processing. [15]

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